

Bao Luu

230 S Chauncey Ave, West Lafayette, IN | 740-334-5150 | luugiabaoluu24@gmail.com | [LinkedIn](#) | github.com/Giabao252

EDUCATION

Purdue University

Expected Graduation: December 2026

Bachelor of Science in Computer and Information Technology

GPA: 3.57/4.0

Relevant Coursework: Data Structures and Algorithms, Data Systems, Object-Oriented Programming, Discrete Mathematics, Computer Systems, Linear Algebra, System Analysis and Design, System Programming, IT Architecture, Software Engineering for Android Apps, Cybersecurity Fundamentals, System Administration, Database Programming, Agentic AI, Applied Generative AI in Mobile Computing

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript/TypeScript, SQL, HTML, CSS, BASH/Shell Scripting

Frameworks & Databases: React.js, Node.js, Nest.js, Express.js, DynamoDB, MongoDB, PostgreSQL, MySQL, SQLite, Git

Cloud & DevOps: AWS (EC2, S3, Lambda, CloudWatch, IAM, VPC), Kubernetes, Docker, GitHub Actions

AI & Automation: TensorFlow, Neural Networks, Large Language Models (LLMs), Anthropic Agent SDK, FastAPI, Qdrant vector database

EXPERIENCE

Software Engineer Intern

September 2024 – February 2026

Mulder Tech Limited (Early-stage startup)

Remote

- Architected and optimized **RESTful APIs** using **Node.js (JavaScript)** and **MongoDB** to handle **100,000+ requests per day** for the startup's core medical traceability product.
- Engineered **Docker** containerized microservices backend architecture with automated **CI/CD pipelines** using GitHub Actions, improving system performance by 40%, reducing deployment time to 2 minutes.
- Designed and implemented **AWS** cloud infrastructure with auto-scaling **EC2** instances, **S3** storage, **VPC** networking with isolated subnets, and **CloudTrail** logging, achieving **99% uptime**.
- Optimized **MongoDB** query performance through compound indexing strategies and aggregation pipeline optimization, reducing average API response times by **60%** and increasing database throughput by **2x**.
- Authored **Postman** API documentation for **80%** of the platform's endpoints with authentication specifications, request/response schemas, and error handling for cross-functional development teams.

High Performance Computing Researcher

February 2025 – August 2025

Purdue's Rosen Center for Advanced Computing

West Lafayette, Indiana

- Managed and troubleshooted **500+ Linux-based** high-performance computing cluster nodes to support **2,000+ Purdue researchers**, monitor infrastructure security and identify system vulnerabilities across distributed environments.
- Built and deployed **Python automation scripts** for cluster monitoring, job scheduling, and incident response workflows, reducing manual intervention by **75%** across Anvil supercomputing nodes.

Software Engineer Intern

May 2024 – August 2024

FPT Software – Financial Services Group

Hanoi, Vietnam

- Collaborated with 12-member cross-functional **Agile/Scrum** team to develop a partner's banking platform using **Nest.js TypeScript, React.js**, and **MySQL** for **300,000+ users**.
- Implemented **multi-factor authentication (MFA)** feature for core banking transactions with OTP verification, reducing unauthorized access attempts by **98%**.
- Optimized MySQL data schemas with relational mappings and indexed queries to improve **data retrieval performance by 20%**.

PROJECTS

Stock Research AI Agent | *Python, FastAPI, MCP, Claude Agent SDK, Qdrant, Cohere, React.js*

May 2026

- Architected an end-to-end **AI platform from ingestion to enrichment, serving, and observability** using Anthropic's Claude Agent SDK with multi-tool orchestration across **2 custom MCP servers** and a **RAG pipeline** over SEC EDGAR filings.
- Built a **RAG retrieval pipeline** combining BM25, Qdrant vector search, and Cohere reranking, evaluated against a 25-question golden dataset for measurable retrieval quality improvements.
- Designed **automated eval harness with DeepEval and GitHub Actions CI/CD**, enforcing more than 90% citation accuracy as a prompt categorization and response quality monitoring system.
- Instrumented production agent with **OpenTelemetry and Grafana Cloud** to monitor **LLM latency**, MCP tool call performance, and retrieval time across live runs.